

```
import cv2
import urllib.request
import numpy as np
from google.colab.patches import cv2_imshow

# Download a sample image automatically
url = "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg"
urllib.request.urlretrieve(url, "sample.jpg")

# Read the image
img = cv2.imread("sample.jpg")

# Define 3 points in the original image
pts1 = np.float32([[50, 50],
                  [200, 50],
                  [50, 200]])

# Define their new positions
pts2 = np.float32([[10, 100],
                  [200, 50],
                  [100, 250]])

# Create the affine transformation matrix
matrix = cv2.getAffineTransform(pts1, pts2)

# Apply affine transformation
rows, cols = img.shape[:2]
affine_img = cv2.warpAffine(img, matrix, (cols, rows))

# Display images
print("Original Image")
cv2_imshow(img)

print("Affine Transformed Image")
cv2_imshow(affine_img)

# Save the output image
cv2.imwrite("affine_transformed.jpg", affine_img)

print("Affine Transformation Completed.")
```

Original Image



Affine Transformed Image

